Nanotechnology - NUIST

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Nanotechnology - NUIST (A35632)

Short Title:	Nanotechnology (NUIST)		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Discipline Specific Technologies		
Author	KOMAHONEY		
Credits	5	**Level:**	Postgraduate

Description of Module / Aims

This module provides an engineering perspective on the interdisciplinary advances in nanotechnology. The module is divided in to five sections, (1) Social Perspectives, (2) Electronic Devices (3) Nanotechnology and Materials (4) Nanotechnology and Manufacturing (5) Environmental Nanotechnology. The module aims at student development gained through an understanding of the interdisciplinary nature of nanotechnology, the horizontal affects that nanotechnology is having modern industry and the disruptive advances that are leading social progress. The research informed ethos of the module comprises both student led and tutor led discovery pathways designed to build discipline specific theoretical and application knowledge through the framing of pertinent research questions and framing a structured response through close interaction with the knowledge base.

Programmes

	Certificate in Biomedical Engineering (`WD_CBIOM_MA`)	1 / 1 / M
`NANO-0001`	MEng in Electronic Engineering (`WD_EELEN_R`)	1 / 1 / E
	MEng in Electronic Information Systems (`WD_EELEN_RI`)	1 / 1 / E
`ENGR-0039`	MSc in Innovative Technology Engineering (`WD_CINNT_R`)	1 / 1 / E
`NANO-0001`	PG Dip in Engineering in Electronic Engineering (`WD_EELEN_G`)	1 / 1 / E
`ENGR-0039`	PG Dip in Science in Innovative Technology Engineering (`WD_CINNT_G`)	1 / 1 / E

Indicative Content

- Societal Nanotechnology
- Quantum Structures
- Tools of Nanotechnology (Visualisation and Manipulation)
- Nanoelectronic Devices -- (Organic Electronics and Single Electron Devices)
- Nanostructures and Nanocomposites
- Environmental Nanotechnology (Sensors, Mitigation and Energy)
- Natural Nanotechnology (Evolution, Nature's Laboratory)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Assess the societal implications of nanotechnology as a horizontal technology and its positive current and future disruptive potential.
2. Develop models to describe quantum systems and thus estimate their size property relationship.
3. Evaluate the role of 1-D and 0-D systems and synthesise associated nanostructures for the development of future electronic devices.
4. Apply SEM, SFM and optical technologies to nanoscale characterisation and manipulation.
5. Distinguish the properties of different thin film composites, evaluate methods for the manufacture and characterisation of thin film materials.
6. Describe intricacies involved in Nanometrology and the NIST roadmap for Nanomanufacture.
7. Evaluate modern nanotechnology innovations for environmental monitoring (air, water, soil) and energy harvesting.

Learning and Teaching Methods

- Lecture and Demonstration
- Mini Project

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	70%	1,2,5,6,7
Project	30%	3,4

Assessment Criteria

Notes:

- >70% In addition to the criteria set out below: The student will have shown exceptional ability in the interpretation and evaluation of nanotechnology innovations and their application in a societally aware context.
- >60% <70% In addition to the criteria set out below: Through a demonstrated ability to interpret and evaluate some nanotechnology concepts the student will have demonstrated a capacity for research informed learning and the development of life-long learning skills.
- >40% <60% The student will have demonstrated a basic understanding of the interdisciplinary nature of nanotechnology, the horizontal affects that nanotechnology is having modern industry and the disruptive nature of nanotechnology. They will also have demonstrated a highly professional approach to submitting all assignments on time.
- <40% The student has failed to demonstrate a basic understanding of the interdisciplinary nature of nanotechnology, the horizontal affects that nanotechnology is having modern industry and the disruptive nature of nanotechnology and/or failed to demonstrate a professional approach to submitting all assignments on time.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical		
Independent Learning	111	

Essential Material

- Hornyak, G.L. _Introduction to Nanoscience_. .: CRC Press, 2008.
- Rubahhn, H.G. _Basics of Nanotechnology_. USA: Wiley, 2008.

Supplementary Material

- "Nano Hub." www.nanohub.org
- Hornyak, G.L. _Fundamentals of Nanotechnology_. USA: CRC Press, 2009.
- Kuno, M. _Introductory Nanoscience_. USA: Garland Science, 2012.
- Poole, C.P. _Introduction to Nanotechnology_. 1st ed.. USA: Wiley, 2003.

Requested Resources

- Lecture Room: Loose Seated